

**Eduardo Cabrera-Lopez**

**Professor Patricia McManus**

**April 11, 2026**

**ITAI 4374**

## **Exploring Neural Information Processing**

### **Overview**

I investigated how artificial and biological neurons interpret information in this lab. Understanding the parallels and discrepancies between simplified neuron models used in artificial intelligence and actual neural systems in the brain was the goal. I created an artificial neuron with weighted inputs and activation functions, as well as a biological neuron simulation that simulates membrane potential over time. In order to see how information is handled by several neurons, I also built basic networks for both models. These tests helped me understand how biological concepts affect AI design and the shortcomings of existing artificial models.

### **Information Processing in Artificial and Biological Neurons: A Comparison**

Electrical and chemical signals are used by biological neurons to process information. Every neuron has a resting membrane potential and changes its voltage in response to incoming impulses. The neuron produces an action potential, sometimes referred to as a spike, when the membrane potential crosses a particular threshold. The neuron either fires fully or not at all in this behavior, which adheres to the all-or-none concept. In my simulation, I found that while stronger inputs forced the neuron beyond the threshold and resulted in spike-like responses, low input currents only little altered the neuron. This illustrates how biological neurons function over time and rely on both temporal dynamics and signal intensity.

In contrast, artificial neurons use mathematical processes to process information. They take numerical inputs, multiply them by weights, add a bias term, and then run the result via an activation function like tanh, sigmoid, or ReLU. Artificial neurons do not rely on spike behavior or time-dependent membrane modifications like genuine neurons do. Rather, they generate continuous output in real time. varied activation functions produced varied output patterns in my tests, demonstrating how function selection influences how information is converted inside the network.

Although biological and artificial neurons both convert inputs into outputs, their methods are essentially different. Artificial neurons use continuous numerical values to encode information, while biological neurons use spikes and time. Nonetheless, the idea of a threshold or transformation function that controls the output response is shared by both systems.

### **AI Systems Inspired by Biology**

Numerous insights from biological neural systems can enhance artificial intelligence system design. Temporal processing is a crucial component. Over time, biological neurons gather signals and react according to the frequency and timing of inputs. This makes it possible for the brain to efficiently process patterns, sequences, and dynamic situations. AI systems that incorporate time-based processing may perform better in tasks like robotics, speech recognition, and real-time decision-making.

Energy efficiency is another crucial factor. Compared to contemporary AI models, the human brain can carry out intricate calculations with relatively little energy. Biological neurons use sparse and effective signaling to fire only when necessary. More energy-efficient AI systems, particularly for edge computing and mobile applications, may be inspired by this idea.

Additionally, biological systems are very resilient and adaptable. They can adapt to new situations, learn from mistakes, and keep working even when certain parts break. On the other hand, a lot of artificial neural networks rely a lot on rigid structures and big datasets. Researchers may create more adaptable and generalizable AI systems by examining biological learning mechanisms like synaptic plasticity and local learning principles.

### **Artificial Neuron Models' Drawbacks and Potential Improvements**

Artificial neuron models are useful, but they have a number of drawbacks. In the first place, they are far simpler than actual biological neurons. Artificial models lack the intricate structures found in biological neurons, such as dendrites, synapses, and neurotransmitter connections. Artificial neurons can't entirely mimic biological information processing because of this simplification.

Second, explicit temporal dynamics are often absent from artificial neurons. The majority of artificial neural networks provide outputs quickly without modeling changes over time since they process inputs in a feedforward fashion. Because of this, time-dependent processes like spike timing and brain adaptability are challenging for conventional models to portray.

Third, biological systems and artificial neural networks employ quite different learning mechanisms. Backpropagation, which is effective for training but not physiologically realistic, is frequently used in artificial networks. Uncertainty surrounds the local learning principles and feedback mechanisms used by biological systems.

Researchers are investigating novel strategies including neuromorphic computing, spiking neural networks, and more biologically inspired architectures to overcome these

constraints. These methods seek to blend the realism of biological processing with the effectiveness of artificial systems.

### **In conclusion**

In summary, this lab showed that while both biological and artificial neurons contribute to the conversion of inputs into outputs, they do so in essentially distinct ways. Artificial neurons use activation functions and mathematical procedures to produce outputs, whereas biological neurons rely on threshold-based firing and time-dependent electrical impulses. Artificial neurons are useful for machine learning and computing, but they fall short of capturing the intricacy of biological systems. Researchers can create more sophisticated AI systems that are effective, flexible, and able to manage challenging real-world tasks by researching the brain.